

Research Article

Study of prevalence of ABO and Rhesus Blood groups in Voluntary Donors in Tertiary care Hospital Blood Centre of western Maharashtra, India.

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Abstract: Background: The ABO system is the most important blood group system in transfusion therapy and was the first blood group system to be discovered. In modern medicine, the need for blood group frequency and prevalence study is multipurpose, as besides its importance in evolution, their relation to disease and environment is becoming increasingly important. Knowledge of the distribution of pattern of different blood groups is very crucial for blood centres and blood transfusion services, which significantly contribute to the health system of the nation and to maintain the donor registry. **Aim:** The study is aimed to know the frequency of ABO and Rh blood group in a population of western Maharashtra, India which is essential for effective management of blood inventory and to maintain blood donor registry. **Material and method:** The present study was conducted at the Government Medical College and Hospital Blood Centre, Miraj. A total of 13858 blood donors were selected and accepted for blood donation as per the blood Donor selection guidelines of the Drug and cosmetics Act, 1940. **Result:** Out of 13858 donors, 13440 (96.98%) were males and 418 (3.02%) were females. The majority of donors belong to the age group 30 to 40 years. The commonest A B O group present was B+ 29.55% followed by O+ 28.33%, followed by A+ + 27.48%, and AB+ 9.54% while 13155(94.92%) donors were Rh Positive, and 698 (5.03%) donors were Rh Negative. **Conclusion:** Knowledge of the distribution of patterns of different blood groups is very crucial for blood centers and blood transfusion services, which significantly contribute to the health system of the nation. Blood group distribution is important for clinical studies, geographical information and forensic studies in general population. It also helps to maintain the donor registry, which is useful in a blood crisis.

Keywords: Blood donors, A B O, blood groups, Drug and cosmetics Act, 1940, donor registry.

INTRODUCTION

At present 33 blood group systems representing over 300 antigens are listed by the International Society of Blood Transfusion. 1 The ABO system is the most important blood group system in transfusion therapy and was the first blood group system to be discovered. This great contribution to medicine was made by Landsteiner in 1900 when he observed that 'the serum of healthy humans not only has an agglutinating effect on animal blood corpuscles, but also on human blood corpuscles from different individuals'. 2

His extensive research on serology based on simple but strong scientific reasoning led to the identification of major blood groups such as O, A, and B types, compatibility testing, and subsequent practices 1 The Rhesus (Rh) system is the second most important blood group system from a transmission point of view. Identification of the Rh system is important to prevent the erythroblastosis Fetalis, which commonly arises when a Rh-negative mother carries Rh Rh-positive fetus. .3

The distribution of A B O and Rh blood groups is important for the effective management of Blood banks.4 The frequency of A B O and RhD phenotypes in different populations has been extensively studied. Different blood

groups are particularly associated with different diseases as well. In modern medicine, the need for blood group frequency and prevalence study is multipurpose, as besides its importance in evolution, its relation to disease and environment is becoming increasingly important. Blood group antigens can also be utilized in genetic research, forensic pathology, Anthropology, and in tracing the ancestral relation of human beings. 5

Ideally, a blood transfusion system should be capable of addressing 100% of clinical demand arising in healthcare facilities within its catchment area. Yet, in many developing and underdeveloped countries, there is a widespread shortfall between the demand and supply of blood due to several barriers. The major factors are increasing requirement for blood and blood products, poor implementation of voluntary donation and blood safety programs in countries, inadequate voluntary non-remunerated blood donation (VNRBD), suboptimal component separation, inadequate infrastructure, equipment and trained human resources, inappropriate use of blood and blood components, poor quality management systems, poor supply chain management systems, lack of cold chain, wastages and expiry of blood (6) In addition to their significance in blood transfusion practice, ABO grouping and Rh typing are also useful in

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clinical studies, genetic studies in population and in solving medico-legal issues. (7)

Knowledge of the distribution of pattern of different blood groups is very crucial for blood centre and blood transfusion services which are significantly contribute to the health system of the nation and to maintain donor registry. Hence this study was conducted to determine the frequency of ABO and Rhesus (Rh) blood groups in a tertiary care Government medical college and hospital Blood Centre in western Maharashtra, India.

Aims and Objectives

To know the frequency of ABO and Rh blood group in a population of Western Maharashtra which is essential for effective management of blood inventory and to maintain blood donor registry.

MATERIALS AND METHODS

- Study setting and duration: The present study was conducted at the Government Medical College and Hospital blood centre, Miraj, located in Western Maharashtra, India.
- Study design and sample size: It is a cross-sectional retrospective study with the sample size of 13858 healthy donors.
- Inclusion criteria: All the voluntary non-remunerated blood donors coming to blood bank and outside the blood bank in voluntary blood donation camps were

eligible as per the guidelines of Drug and cosmetics Act, 1940

- Exclusion criteria: Donors don't qualify as per the guidelines of Drug and cosmetics Act, 1940

Data collection:

Information regarding personal details, demographic details, occupation, and past medical history was elicited. Individuals with good health, mental alert, physical fit were selected as blood donors. The donors were then asked to sign the donor questionnaire as well as donor register inclusive of the inform consent.

After blood donation, A B O and Rh typing were done on each sample using the tube agglutination method according to SOPs. The Monoclonal type Antisera reagents anti-A and anti-B, and anti-D, manufactured by Tulip Diagnostics, were used. O blood group donors were tested by anti-H Lectin to rule out the Bombay blood group. Each new lot of Antisera was used after a quality control test, like Titer and avidity. Also, on each day, a daily quality control test is also performed before use of Antisera.

Reverse grouping was done using known pooled A cells, B cells and O cells from three different known donor samples. The final blood group is confirmed only if both Forward group and Reverse group are identical. All the week D groups were considered positive.

RESULTS

Table 1: Distribution of donors according to Gender.

Gender	Number of donors	Percentage
Male	13440	96.98%
Female	418	3.02%
Total	13858	100%

Figure-1: Distribution of donors according to ABO blood groups.

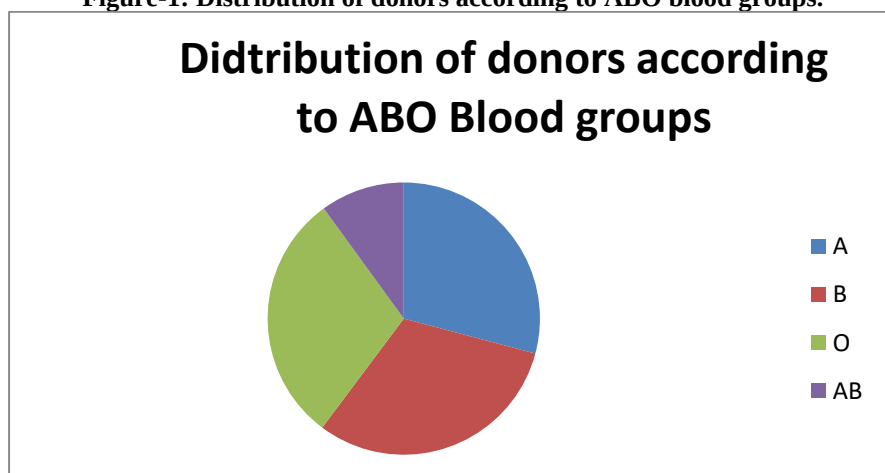


Table 2: Distribution of donors according to ABO and Rh blood group system

Types of Blood Group	Number of donors	Percentage
A Rh -Positive	3809	27.48%

A Rh -Negative	226	1.63%
B Rh -Positive	4096	29.55%
B Rh -Negative	213	1.53%
O Rh -Positive	3927	28.33%
O Rh -Negative	200	1.44%
AB Rh -Positive	1323	9.54%
AB Rh -Negative	64	0.46%
Total	13858	100%

Table 3: Comparative studies with this study

Study	Place	No. studied	ABO blood group				Rh blood group	
			A	B	AB	O	Rh D Positive	Rh D Negative
Studies from India								
Periyavan et al ⁸	Bengaluru	36,964	23.9	29.9	6.4	39.8	94.2	5.8
Das et al ⁹	Vellore	150,536	21.8	32.7	6.7	38.8	94.5	5.5
Mallikarjuna et al ¹⁰	Devanagere	19,413	26.1	29.9	7.2	36.8	94.5	5.5
Girish et al ¹¹	Shimoga Malnad	ND	24.3	29.4	7.1	39.2	94.9	5.5
Chandra et al ¹²	Lucknow	1,40,320	21.7	39.8	9.3	29.1	95.7	4.3
Sidhu et al ¹³	Punjab	1,150	21.9	37.6	9.3	31.2	97.3	2.7
Patel et al ¹⁴	Western Ahmedabad	5,316	21.9	39.4	7.9	30.8	95.1	4.9
Wadhwa et al ¹⁵	eastern Ahmedabad	ND	23.3	35.5	8.8	32.5	94.2	5.8
Mehta et al ¹⁶	Surat	ND	24.1	34.9	8.7	32.3	94.2	5.8
Gupta et al ¹⁴	Indore	17,080	24.2	35.2	9.1	31.5	95.4	4.6
Naga et al ¹⁸	Durgapur	3850	23.9	33.6	7.7	34.8	94.7	5.3
Present Study	Western Maharashtra	13858	21.1	40.8	7.6	30.5	91.4	8.6
Studies from other parts								
Anees et al ¹⁹	Punjab, Pakistan	2542	34	29	4	33	96.7	3.3
Pramanik et al ²⁰	Kathmandu, Nepal	120	26.2	22	4.4	47.5	80.3	19.7
Lyco et al ²¹	Nairobi, Kenya	38898	20.0	32.2	6.1	41.7	92.8	7.2

Table 4: Distribution of donors according to Rhesus Phenotype.

Blood Group	Total Number of donors	Male donors	Female donors
Rh -Positive	13155(94.92%)	12,900(93.08%)	255(1.84%)
Rh -Negative	703(5.08%)	540(3.89%)	163(1.19%)

DISCUSSION

All human populations share the same blood group systems, although they differ in the frequency of specific types. The frequencies of ABO and Rh blood groups vary from one population to another and time to time in the same region.

Current study shows that most donors were male (96.98%) Table 1, which is consistent with other studies in most regions globally. The lower rate of donation in females (3.02%) is due to fear of donation, iron deficiency anemia and postpartum blood loss which leads to anemia in most of female donors.

In studies spanning diverse cultural and geographic groups, the most common age range for blood donation is 18 to 44 years. Our study results are consistent with this Global trend. The improved interest and ability among younger adults to donate maybe related to awareness, better Physical health and greater mobility. Older individuals may suffer from Medical conditions such as ischemic heart disease, diabetes mellitus, hypertension hence negatively impacting their ability to be well enough to donate blood.

The majority of donors in our study were voluntary, non-remunerated donors, 100%. This is consistent with other studies and global Trends. The donation of blood by voluntary, non-remunerated blood donors is critical for the safety and sustainability of the national blood supply. The national blood donation system in which replacements are typically unable to meet clinical demands for blood, while paid family members contributing a serious threat to the health and safety of the recipients. Who's recommendations are to create a health system based 100% on voluntary donation

Table 3 shows that Frequency of A B O blood groups of the present study was compared to the studies done at Lucknow by Chandra et al 12, at Punjab Sidhu et al¹³ at Ahmedabad by Patel PA et al¹⁴ and Wadhwa MK et al 15, at Surat by Mehta T et al 16 and at Indore by Gupta H et al. 17 In all these studies predominant blood group was 'B 'which was followed by O, A and AB

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On the other hand, studies were done at Shimoga Malnad by, at Bangalore by Periyavan A et al 8 Das et al 9, Mallikarjuna et al 10, Girish CJ et al 11 and at Durgapur by Nag I et al 18, the commonest A B O blood group was 'O' which is in contrast to present study where B is the most common A B O blood group

In other countries of the world, like in Pakistan by Anees et al 19, Pramanik et al 20 in Kathmandu, Nepal, Lyco et al 21 in Nairobi, Kenya, showed that the blood group O was highly frequent which contrasts with our presence study.

Table 4 shows that, frequency of Rhesus blood groups of the present study was 94.92% and 5.08% for RH positive and RH negative donors, respectively. Which is like all studies except in the USA, where it is 85% and 15% respectively. Some of the factors which can affect the overall distribution of the ABO Rh blood group depend upon the donor selection and deferral which are unique to specific location.

CONCLUSION

Knowledge of the distribution of patterns of different blood groups is very crucial for blood centers and blood transfusion services, which significantly contribute to the health system of the nation. Blood group distribution is important for clinical studies, geographical information, and forensic studies in the general population.

It also helps to maintain the donor registry, as there are very few voluntary donors in society, of which very few are regular donors. By preparing or making a Registry of these regular Donors as per A, B, and O Blood group differentiations, this Registry will be valuable or of great help during a blood crisis. That is why it is important to maintain the distribution of A, B, O, and Rh Blood groups.

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